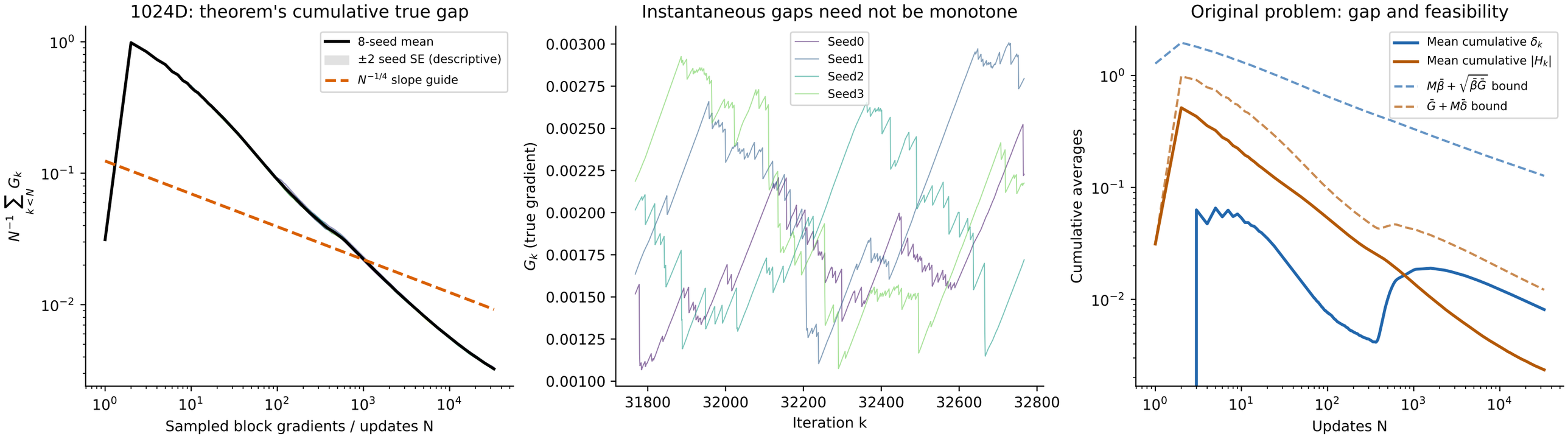

1024D: theorem's cumulative true gap
8-seed mean
±2 seed SE (descriptive)
$N^{-1/4}$ slope guide
$N^{-1}\sum_{k<N} G_k$
Sampled block gradients / updates N
Instantaneous gaps need not be monotone
Seed0
Seed1
Seed2
Seed3
G_k (true gradient)
Iteration k
Original problem: gap and feasibility
Mean cumulative δ_k
Mean cumulative $|H_k|$
$M\bar{\beta} + \sqrt{\bar{\beta}\bar{G}}$ bound
$\bar{G} + M\bar{\delta}$ bound
Cumulative averages
Updates N